

CAREERBRIDGE – DAY 3 EXERCISE Topic: Entity Relationship Diagram (ERD) and Database Relationships PART 1 — IDENTIFY THE ENTITIES

My Proposed Entities

1. User
2. Job seeker
3. Company
4. Job
5. Application
6. Job Category
7. Admin
8. CV

PART 2 — DEFINE THE ATTRIBUTES

Entity 1

Entity Name: User

Attribute	Purpose
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UserID	Uniquely identifies each user
FirstName	Stores user's firstname
Lastname	Stores user's surname
Email	Stores user's email address
Password	Stores user's account password
Role	Identifies whether user is a job seeker, employer or admin
Phone	Stores user's phone number
CreatedOn	Store the date the user created the account

Primary Key: UserID

Possible Foreign Key: None

Entity 2

Entity Name: JobSeeker

Attribute	Purpose
JobSeekerID	Uniquely identifies a Job Seeker
UserID	Links the Job Seeker to their use account
Phone	Stores contact number
Location	Stores the Job Seeker's location
Skills	Stores the Job Seeker's skills
Education	Stores educational information

Primary Key: JobSeekerID

Possible Foreign Key: UserID

Entity 3

Entity Name: Admin

Attribute	Purpose
AdminID	Uniquely identifies an administrator
UserID	Links an admin to a user account
AdminRole	Defines the administrator's role
CreatedDate	Records the date the admin account was created
Status	Shows whether an admin is active or not

Primary Key: AdminID

Possible Foreign Key: UserID

Entity 4

Entity Name: Company

Attribute	Purpose
CompanyID	Uniquely identifies each company

CompanyName	Stores the name of the company
Email	Stores the company's email
Phone	Stores the company's phone number
Address	Stores the company's address
Description	Provides information about the company
CompanyWebsite	Stores the link to the company's website
CreatedDate	Stores the date the company profile was created
Status	Stores whether the company is active or not

Primary Key: CompanyID

Possible Foreign Key: None

Entity 5

Entity name: Job

Attribute	Purpose
JobID	Uniquely identifies a job
CompanyID	Identifies the company that posted the job
CategoryID	Identifies the job category
JobTitle	Stores the title of the job
Description	Stores details about the job
Location	Stores the job's location
JobType	Stores the job's type
PostedDate	Stores the date the job was posted

ApplicationDeadline	Stores the deadline of the application
Status	Shows whether the job is active or not
Primary Key: JobID	
Possible Foreign Key:	
• CompanyID • CategoryID	

Entity 6

Entity Name: Application

Attribute	Purpose
ApplicationID	Uniquely identifies an application
JobSeekerID	Identifies the person who applied
JobID	Identifies the job being applied for
ApplicationDate	Records when the application was submitted
Status	Shows the application status
CoverID	Locates the CV
Primary Key: ApplicationID	
Possible Foreign Key:	
• JobSeekerID • JobID • CVID	

Entity 7

Entity Name: Job Category

Attribute	Purpose
CategoryID	Uniquely identifies a category

CategoryName	Stores the category name
Description	Describes the category
Attribute	Purpose
Status	Shows whether the category is active

Primary Key: CategoryID

Possible Foreign Key: None

Entity 8

Entity Name: CV

Attribute	Purpose
CVID	Uniquely identifies a CV
JobSeekerID	Identifies the owner of the CV
CVTitle	Stores the CV title
FilePath	Stores the location of the CV
UploadDate	Stores the date it was uploaded
LastUpdated	Stores the date it was last updated

Primary Key: CVID

Possible Foreign Key: JobSeekerID

PART 3 — PRIMARY KEYS AND FOREIGN KEYS

Entity	Primary Key	Foreign Key(s)
User	UserID	None
JobSeeker	JobSeekerID	UserID
Admin	AdminID	UserID
Entity	Primary Key	Foreign Key(s)
Company	CompanyID	None

Job	JobID	CompanyID, CategoryID
Application	ApplicationID	JobSeekerID, JobID , CVID
Job Category	CategoryID	None
CV	CVID	JobSeekerID

PART 4 — DEFINING THE RELATIONSHIPS

Relationship 1

A company can post several jobs.

A job belongs to one company.

Relationship: One-to-Many

Explanation: One company can post different types of jobs but those jobs are being posted by just one company.

Relationship 2

A Job Seeker can submit several applications.

Each application belongs to one Job Seeker.

Relationship: one-to-Many

Explanation: One job seeker can apply for several jobs.

Relationship 3

A job can receive several applications.

Each application belongs to one job.

Relationship: One-to-Many

Explanation: Different people can apply for a single job but all those applications are sent for that particular job.

Relationship 4

A job belongs to one job category.

A job category may contain several jobs.

Relationship: One-to-Many

Explanation: A job can only belong to one category, but that same category may contain different types of jobs.

Relationship 5

A user may have one Job Seeker profile.

Relationship: One-to-One

Explanation: A job seeker can have only one profile

PART 5 — THINK ABOUT MANY-TO-MANY RELATIONSHIPS

Consider this situation:

A Job Seeker can apply for many jobs.

A job can have many Job Seekers applying for it.

Answer the following:

Question 1

Is the relationship between Job Seeker and Job:

☐ One-to-One

☐ One-to-Many

☐ Many-to-Many

Question 2

Which entity can be used to resolve this relationship?

Application

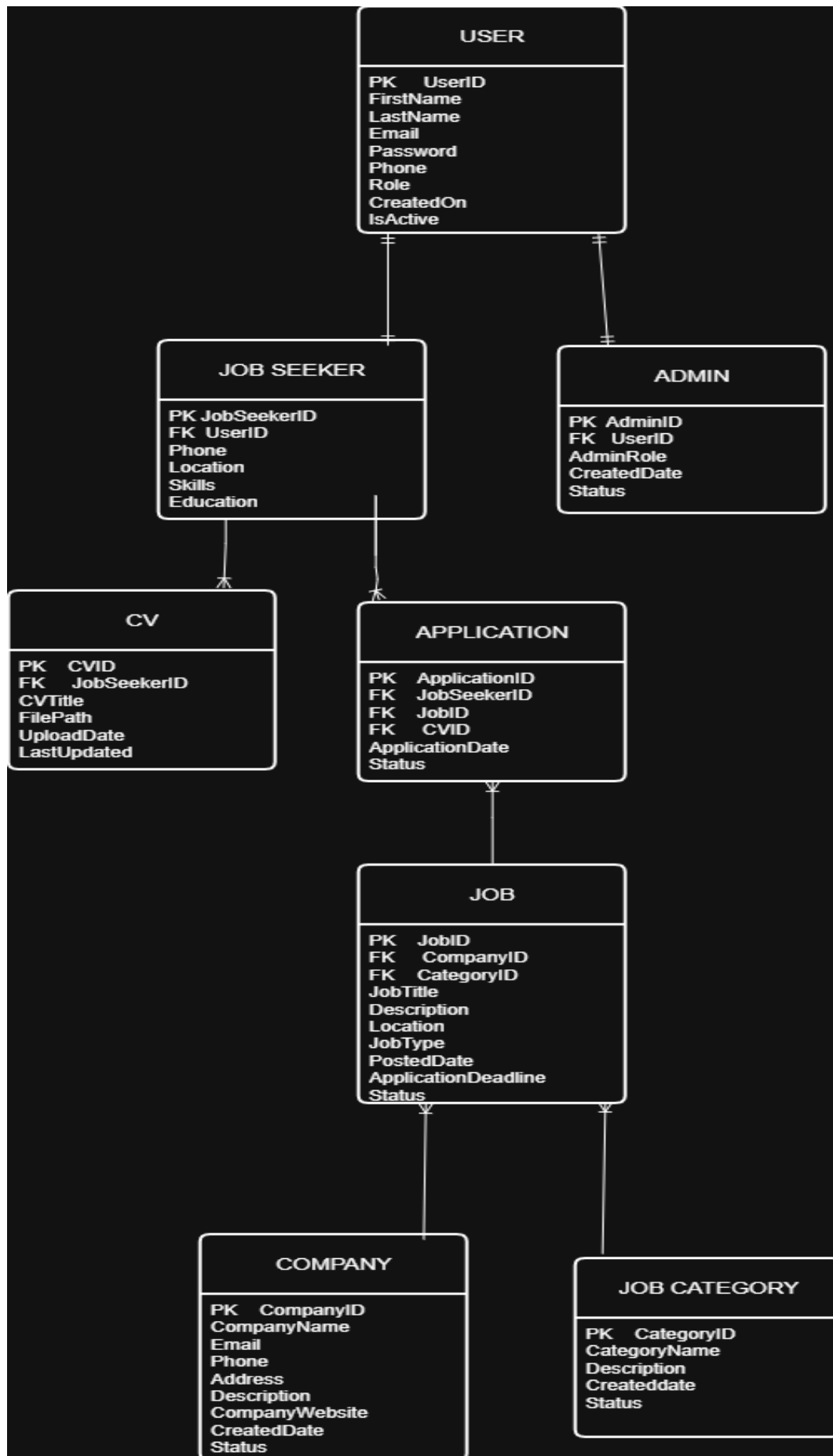
Question 3

Which two Foreign Keys should exist inside that entity?

1. JobSeekerID

2. JobID

PART 6 — ERD



PART 7 — BUSINESS RULES

1. One job belongs to one category, but each category can contain different jobs.
2. One User may have one Job profile.
3. One user may have one Admin profile.
4. One job seeker can apply for different jobs.
5. One job can receive different applications, but each application belongs to one job.
6. One job seeker can have several CVs, but each CV belongs to only one job seeker.
7. One job seeker can apply for different jobs.
8. A company can have only one profile.

PART 8 — THINKING QUESTIONS

1. What is an Entity?

An entity is a major object in a system whose information needs to be stored.

2. What is a Primary Key?

A primary key is an attribute that uniquely identifies a record in a table

3. What is a Foreign Key?

A foreign key is a key that belongs to another table and links those two tables.

4. Why do we use relationships between tables?

Relationships allow related information to be connected while keeping data organized in separate tables

5. Why should CompanyName not be repeated inside every Job Application record?

Because it will create unnecessary duplicate data.

6. Why is the Application entity important?

It is important because it links the job seeker to the job he or she has applied for.

7. What could happen if Foreign Keys are not used correctly?

The database will contain incorrect data.

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